

Predicting Future California Dashboard Performance Using Machine Learning

Aspire Public Schools Dashboard Classification Analysis

Kate Hill
Springboard Data Science Career Track
Capstone Two Final Report

Executive Summary

This project explored whether historical California School Dashboard indicators could be used to predict future Dashboard color classifications for Aspire Public Schools. The California Dashboard serves as the state accountability system for measuring school performance across indicators such as academic achievement, chronic absenteeism, suspension rates, and college and career readiness.

Using historical Dashboard data from Aspire Public Schools, machine learning classification models were developed to predict the following year's Dashboard color classification. The project included data wrangling, exploratory data analysis, feature engineering, preprocessing, and predictive modeling.

Three machine learning models were evaluated:

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

After comparing model performance, the Gradient Boosting model was selected as the final model due to its ability to capture nonlinear relationships and provide more balanced multiclass classification performance.

Although overall predictive accuracy remained moderate due to the relatively small dataset and the complexity of multiclass prediction, the results demonstrated that historical Dashboard indicators contain meaningful predictive information regarding future accountability outcomes.

This project demonstrates how predictive analytics and machine learning techniques may support proactive educational planning, early intervention efforts, and data-informed decision-making within school systems.

Problem Statement

Educational organizations frequently rely on historical accountability data to evaluate school performance and identify schools requiring additional support. However, Dashboard results are typically released after performance trends have already occurred, limiting opportunities for proactive intervention and planning.

The purpose of this project was to determine whether machine learning models could predict future California Dashboard color classifications using historical Dashboard performance indicators from Aspire Public Schools.

If future Dashboard outcomes can be predicted with reasonable accuracy, educational organizations may be able to:

- identify schools requiring intervention earlier,
- allocate support resources more strategically,
- and proactively respond to declining performance trends before official Dashboard release cycles.

This project specifically focused on predicting the following year's Dashboard color classification using historical school-level Dashboard indicators.

Data Overview

The dataset used in this project consisted of historical California School Dashboard data for Aspire Public Schools.

The dataset included:

- Aspire schools only,
- All Students subgroup only,
- one row per school × indicator × year,
- and cleaned, standardized Dashboard performance variables.

The dataset contained 427 observations across multiple Dashboard indicators, including:

- chronic absenteeism,
- English Language Arts (ELA),
- mathematics,
- and suspension rate.

The target variable for the project was the future Dashboard color classification, representing the following year's accountability outcome for each school and indicator combination.

Because Dashboard colors represent categorical performance levels, the project was structured as a multiclass classification problem.

Data Wrangling

Several data wrangling and preparation steps were required before modeling could begin.

The original Dashboard files were combined into a unified dataset containing multiple years of school-level accountability data. Non-Aspire schools and irrelevant records were removed to focus the analysis specifically on Aspire Public Schools.

Additional cleaning steps included:

- removing unnecessary columns,
- handling missing values,
- standardizing variable formats,
- and preparing temporal relationships between historical and future Dashboard outcomes.

Feature engineering was also performed to create the future prediction target variable. Historical school performance indicators were shifted forward to allow the models to predict next year's Dashboard color classification using prior-year Dashboard performance metrics.

The dataset was then prepared for modeling by:

- encoding categorical variables,
 - creating dummy variables,
 - standardizing numeric features,
 - and splitting the data into training and testing datasets.
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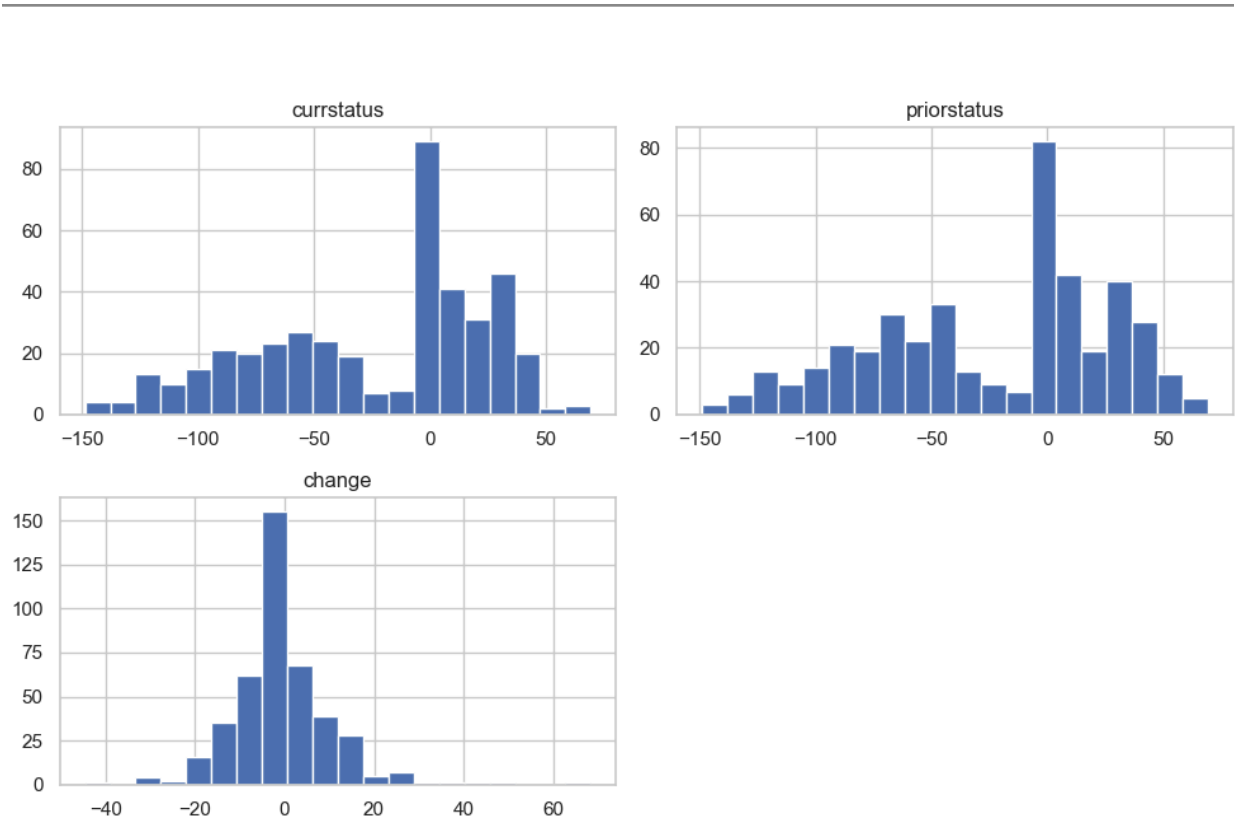
Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to better understand the structure of the dataset and identify relationships among Dashboard indicators and future performance outcomes.

The analysis focused on:

- feature distributions,
- temporal relationships,
- subgroup differences,
- and correlations between Dashboard variables and future Dashboard color classifications.

Several visualizations were created to examine feature distributions and identify potential predictive relationships within the data.



(Distribution Histograms Image)

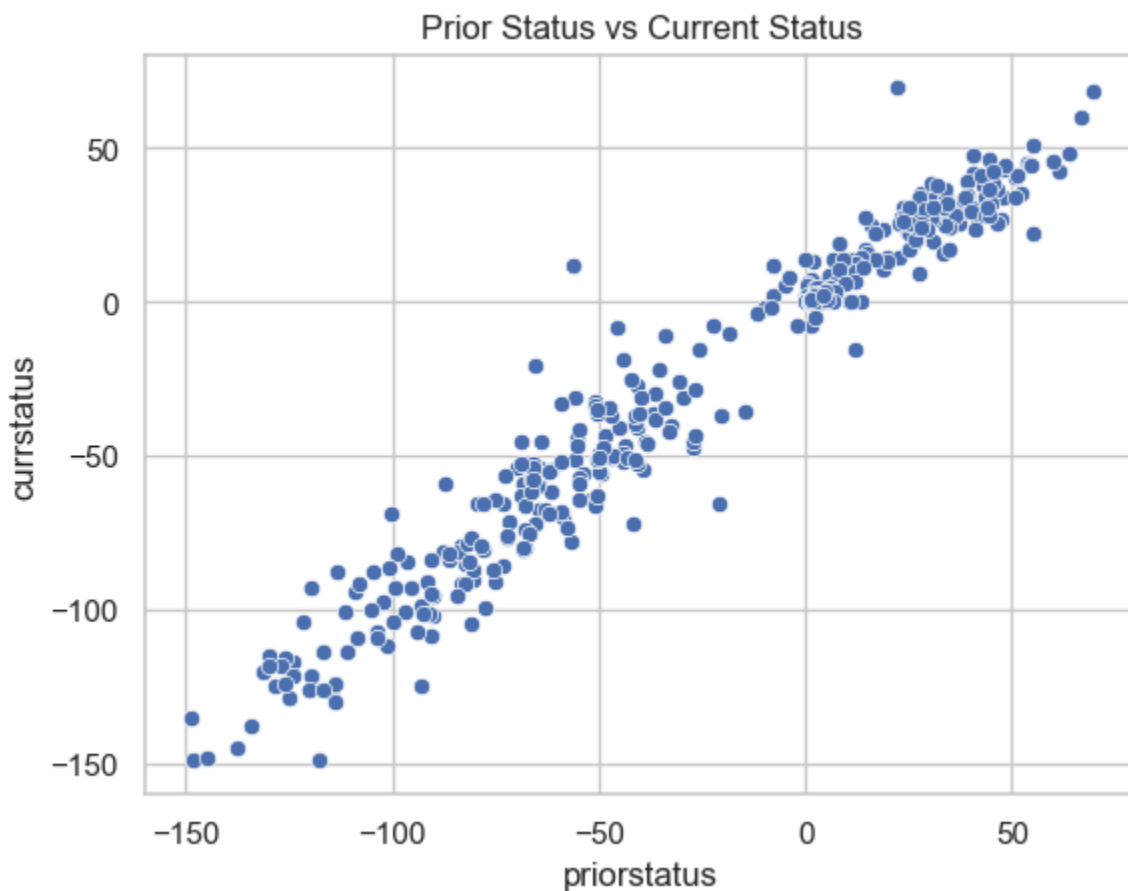
Figure 1: Distribution of Dashboard Performance Variables

Figure 1 displays the distributions of current status, prior status, and year-to-year change variables across Aspire Public Schools Dashboard records.

Current status and prior status demonstrate broad variation in school performance outcomes, while the change variable is more tightly centered near zero. This suggests that although school performance varies substantially across Dashboard indicators, most schools experience relatively modest annual performance shifts.

The similarity between current and prior status distributions also suggests strong temporal consistency in Dashboard performance outcomes, indicating that historical accountability data may provide meaningful predictive signal for forecasting future Dashboard classifications.

To further evaluate relationships between historical performance and current outcomes, scatterplots were used to examine relationships between prior status, annual change, and current status.





(Scatterplot Image)

Figure 2: Relationships Between Prior Status, Change, and Current Status

Figure 2 illustrates the relationships between prior status, annual performance change, and current Dashboard status.

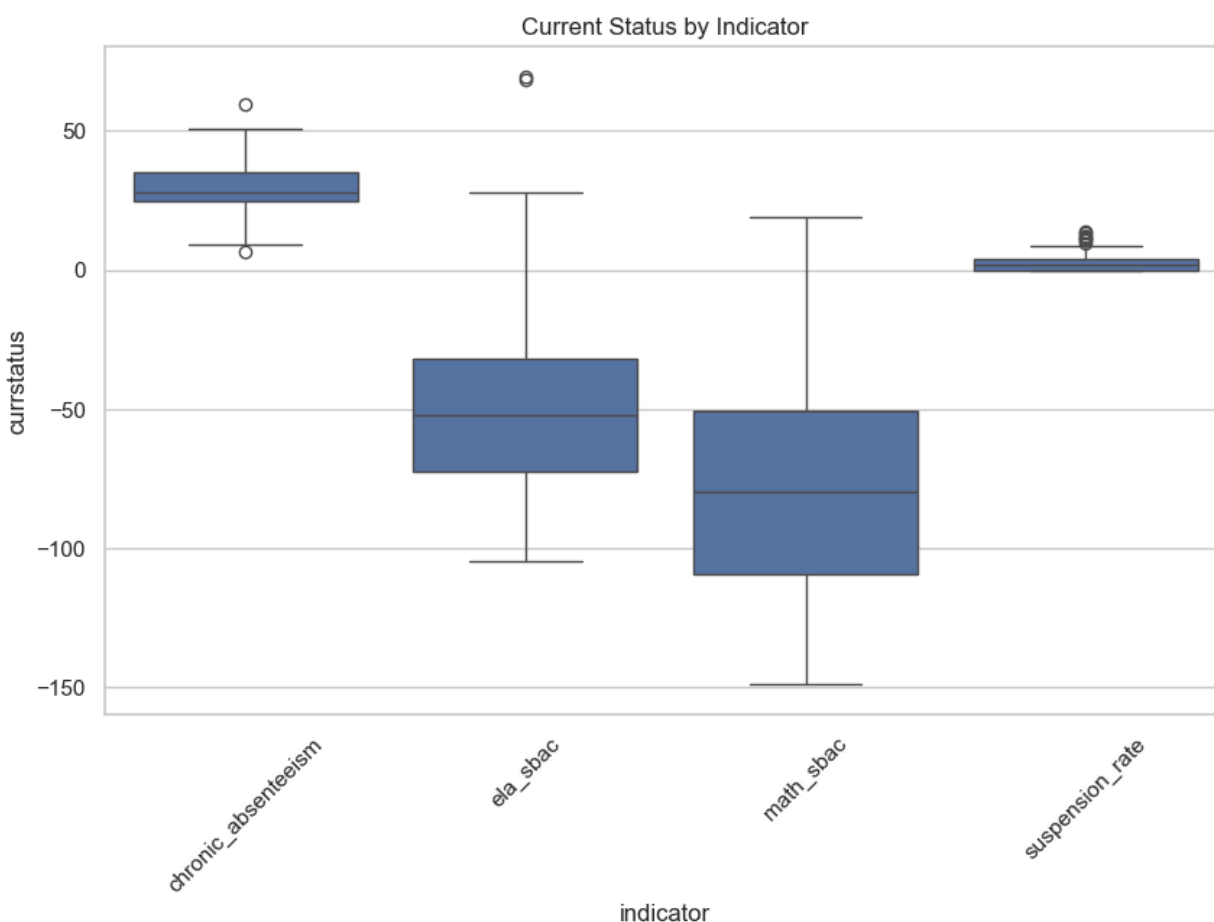
The relationship between prior status and current status demonstrates a strong positive linear trend, indicating substantial year-to-year stability in school accountability performance. Schools with stronger prior performance generally continued to demonstrate stronger current performance outcomes.

In contrast, the relationship between annual change and current status appears more dispersed, suggesting that year-to-year performance shifts add variability but are less directly predictive than historical status measures alone.

These findings support the hypothesis that historical Dashboard performance contains meaningful predictive information for forecasting future Dashboard outcomes.

Additional analysis was conducted to determine whether Dashboard performance distributions differed across indicator categories.

Because the California Dashboard combines multiple educational accountability measures, it was important to evaluate whether indicator type itself contributed meaningful variation to the prediction problem.



(Current Status by Indicator Boxplot)

Figure 3: Current Status by Dashboard Indicator

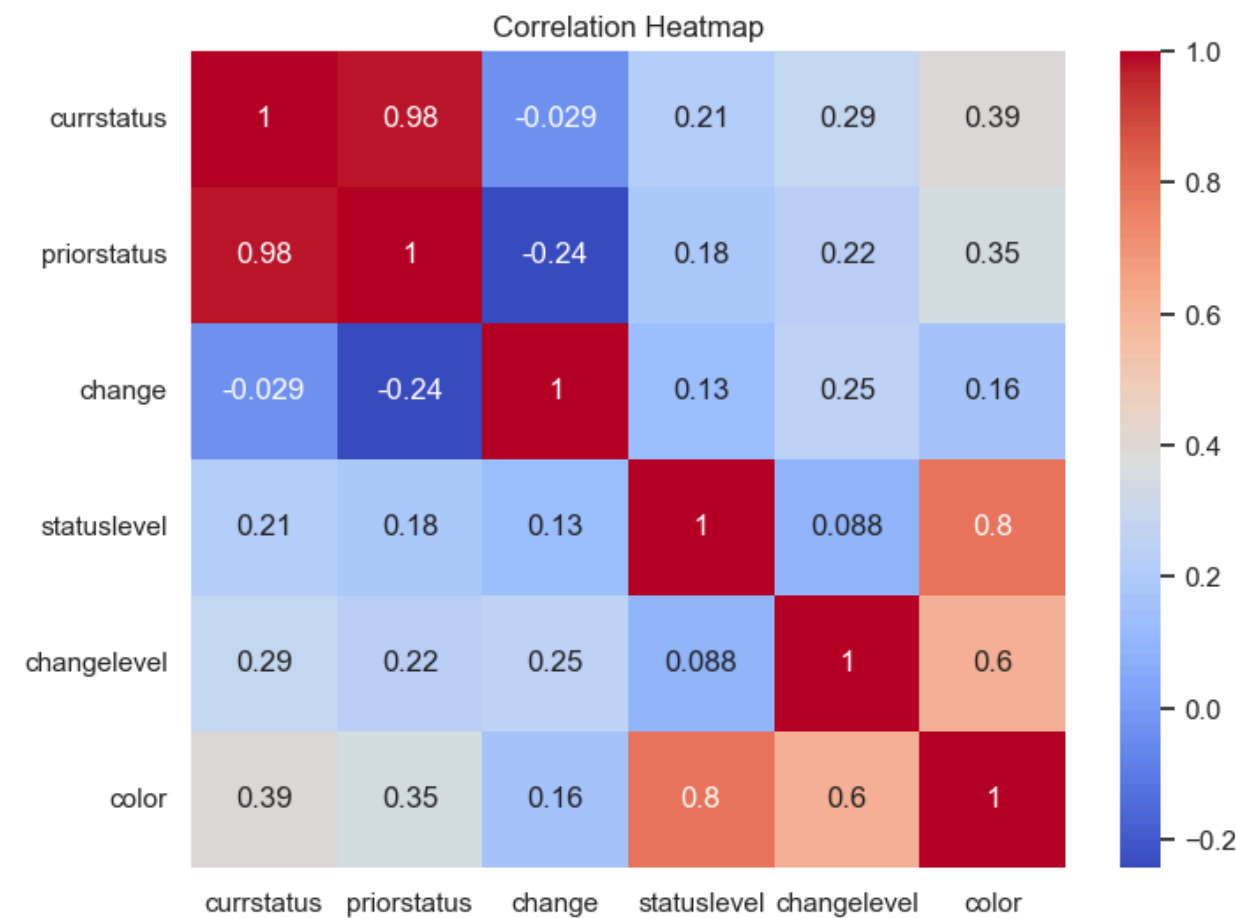
Figure 3 displays the distribution of current status values across Dashboard indicator categories.

Substantial variation exists among indicators. Chronic absenteeism demonstrated higher status values overall, while ELA and Mathematics indicators showed lower and more dispersed performance distributions.

Suspension rate indicators were tightly clustered near zero, suggesting less variability relative to academic indicators.

These differences indicate that indicator category is an important feature within the modeling process and may contribute meaningful predictive information when forecasting future Dashboard classifications.

To further examine relationships among Dashboard variables, a correlation heatmap was created to quantify the strength of associations between accountability indicators and Dashboard color classifications.



(Correlation Heatmap Image)

Figure 4: Correlation Heatmap of Dashboard Variables

Figure 4 illustrates correlations among key Dashboard performance variables.

Current status and prior status demonstrated an extremely strong positive correlation (0.98), confirming the strong temporal stability observed in earlier visualizations.

Status level and change level also showed strong relationships with Dashboard color classifications, which is consistent with the California Dashboard accountability framework.

The heatmap further demonstrates that multiple Dashboard indicators contain meaningful predictive signal related to future accountability outcomes, supporting the use of machine learning classification models for forecasting Dashboard performance.

Preprocessing and Feature Engineering

Before training machine learning models, several preprocessing and feature engineering steps were completed.

Categorical variables were converted into numeric representations using dummy variable encoding. Additional engineered features included:

- numeric year conversion,
- indicator encoding,
- and a binary “improved” feature representing positive year-over-year performance change.

Numeric features were standardized to place variables on comparable scales and improve model performance across different machine learning algorithms.

The dataset was then divided into:

- training data,
- and testing data.

The training dataset contained 213 observations, while the testing dataset contained 71 observations.

Modeling

Three classification models were developed and evaluated during this project:

1. Logistic Regression
2. Random Forest Classifier
3. Gradient Boosting Classifier

The models were selected to compare:

- linear modeling approaches,
- ensemble tree-based methods,
- and boosting techniques capable of capturing nonlinear relationships.

Model performance was evaluated using:

- accuracy,
 - precision,
 - recall,
 - F1-score,
 - and confusion matrices.
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Logistic Regression Results

The Logistic Regression model achieved an accuracy of approximately 36.6% on the testing dataset.

The model performed best when predicting Dashboard color class 3 but struggled to accurately classify several smaller Dashboard categories. Because Logistic Regression primarily models linear relationships, it had difficulty capturing the complexity of educational accountability outcomes.

Despite these limitations, Logistic Regression served as an important baseline model for comparison against more advanced ensemble methods.

Random Forest Results

The Random Forest Classifier achieved an accuracy of approximately 33.8%.

Although Random Forest can model nonlinear relationships and feature interactions, the relatively small dataset size may have limited the model's ability to generalize effectively.

The model again performed best on Dashboard color class 3 while struggling with several minority classes.

Despite lower overall accuracy, Random Forest remained valuable because it captured more complex feature interactions than the baseline linear model.

Gradient Boosting Results

The Gradient Boosting model achieved an accuracy of approximately 36.6%, matching Logistic Regression while demonstrating more balanced multiclass classification performance across Dashboard categories.

Gradient Boosting performed particularly well because it can:

- capture nonlinear relationships,
- reduce prediction bias,
- and model complex interactions among educational indicators.

Because of its balanced performance and flexibility, Gradient Boosting was selected as the final model for the project.

Model Comparison

Model	Accuracy
Logistic Regression	36.6%
Random Forest	33.8%
Gradient Boosting	36.6%

While overall model accuracy remained moderate, the results suggest that historical Dashboard indicators contain meaningful predictive information regarding future accountability outcomes.

The relatively modest performance across models is likely influenced by:

- limited dataset size,
 - multiclass prediction difficulty,
 - overlapping Dashboard categories,
 - and the inherent complexity of educational performance systems.
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INSERT FIGURE 5 HERE

(Model Accuracy Comparison Bar Chart)

Figure 5: Model Accuracy Comparison

Figure 5 compares model performance across the three machine learning classification algorithms evaluated during the project.

Logistic Regression and Gradient Boosting achieved the highest overall accuracy scores, while Random Forest performed slightly lower on the testing dataset.

Although overall accuracy remained moderate, the results suggest that historical Dashboard indicators contain measurable predictive signal related to future accountability outcomes.

Final Model Selection

The final model selected for this project was the Gradient Boosting Classifier.

Gradient Boosting was selected because it:

- achieved competitive accuracy,
- captured nonlinear relationships,
- handled multiclass prediction effectively,
- and demonstrated more balanced classification performance across Dashboard categories.

Although predictive accuracy was not exceptionally high, the model still demonstrated that machine learning can identify meaningful patterns within educational accountability data.

Recommendations

Based on the findings from this project, the following recommendations are proposed:

1. Use Predictive Modeling as an Early-Warning Support Tool

Machine learning models may help educational organizations identify schools at greater risk of Dashboard performance declines before official Dashboard release cycles.

2. Expand Longitudinal Data Collection

Including additional years of Dashboard data, subgroup-level metrics, and operational variables may improve future predictive performance.

3. Integrate Predictive Analytics Into Continuous Improvement Planning

Predictive models could support proactive intervention planning, resource allocation, and school improvement discussions within educational organizations.

Future Research

Several opportunities exist for expanding this project in the future.

Potential improvements include:

- incorporating additional years of Dashboard data,
- including subgroup-level accountability metrics,
- adding demographic and operational variables,
- performing more advanced hyperparameter tuning,
- and testing additional ensemble machine learning models.

Future work could also explore:

- regionalized modeling approaches,
- temporal forecasting methods,
- and integration with attendance, staffing, and finance datasets.

Conclusion

This project demonstrated how machine learning techniques can be applied to educational accountability data to predict future California Dashboard performance outcomes for Aspire Public Schools.

Three classification models were evaluated:

- Logistic Regression,
- Random Forest,
- and Gradient Boosting.

After comparing model performance, Gradient Boosting was selected as the final model due to its ability to model nonlinear relationships and provide balanced multiclass classification performance.

Although prediction accuracy remained moderate, the results suggest that historical Dashboard indicators contain meaningful predictive information about future accountability outcomes.

This project highlights the potential for predictive analytics to support proactive educational planning, continuous improvement efforts, and data-informed decision-making within school systems.